

Lecture 2: Meningitis & Encephalitis II

Diagnosis of a case of meningitis

➤ Clinical manifestation

Start Empiric Antibiotic----- **Medical emergency*******

Meningitis can be a life-threatening emergency, especially if a person has bacterial meningitis

➤ Laboratory diagnosis

□ *CSF Examination:*

The most reliable way to diagnose meningitis is with a lumbar puncture. During a lumbar puncture, a doctor inserts a long, thin needle in between two vertebrae in the lower back. This allows them to withdraw some cerebrospinal fluid (CSF

1. Sample collection “Lumbar puncture”

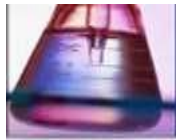
2. Transport “within 1 h”

3. Processing of sample:

□ Physical examination “clear or turbid, normal or under pressure

□ Chemical examination “glucose, protein”

□ Cytological examination “WBCs, RBCs”



Cell count & Biochemical

Diagnosis	Total Cells (per mm ³)	Predominant cell	Proteins (mg/dL)	Glucose (mg/dL)	CSF Pressure (mm Hg)
Normal CSF	0-5	Lymphocyte	15-45	40-80	70-180
Pyogenic Meningitis	↑↑↑ed (5-20,000)	Polymorph (Neutrophil)	↑↑↑ed (>100)	↓↓ed	↑ed
Tubercular Meningitis	↑ed (5-2,000)	Lymphocyte	↑↑ed (>45)	↓ed	↑ed
Viral Meningitis	↑ed (5-2,000)	Lymphocyte	↑ed (>45)	Normal	↑ed or Normal

□ Microbiological

examination Smear 'Gram & Zeihl

Nelseen stain

Culture & sensitivity 'blood agar- chocolate agar....

Identification biochemical tests

Rapid detection tests for antigen "latex agglutination" for meningococci, pneumococci, H influenzae B

Molecular diagnosis: PCR

Indirect Serological diagnosis for detection of antibodies.

ESR, CRP, CBC, Blood culture, imaging....

Encephalitis

Diffuse inflammation of brain parenchyma.

Meningoencephalitis

Inflammation of brain parenchyma and meninges.

□ Causes of Encephalitis:

□ **Viral Commonest**

□ **Non-Viral: Rare**

Bacterial “TB, Syphilis, ..

Fungal “Cryptococcus neoformans, ...

Parasites “Malaria, Toxoplasmosis.....

□ Causes of Viral Encephalitis:

1. Herpes viruses: Herpes simplex^{1,2} Varicella zoster, Cytomegalovirus, Epstein-Barr virus
2. Enteroviruses: Poliovirus
3. Rabies
4. Childhood viruses: Measles, Mumps....
5. Arboviruses “Arthropod borne viruses”: West Nile Virus, Japanese encephalitis “Geographic distribution”

□ **Pathogenesis of Viral Encephalitis:** depend on the virus

- Primary: direct viral destruction of the cells
- Secondary: para or post-infectious inflammation or immune- mediated response.

Most viruses primarily infect brain parenchyma and neural cells.

Some cause vasculitis and Demyelination may follow infection

□ Mode of infection:

- Herpes Simplex viruses: person to person contact
- Enteroviruses: contact with body fluids
- HIV: infected blood or body fluids to blood stream
- Arboviruses: bite of arthropods as mosquitos that pick virus from infected host animal
- Rabies: bite or saliva of infected animals

□ Clinical picture of Encephalitis

Typical presentations:

- Acute flu-like prodrome

- High fever, severe headache
- Altered consciousness (lethargy, drowsiness, confusion, coma)
- Seizures
- Focal neurological

signs Less common

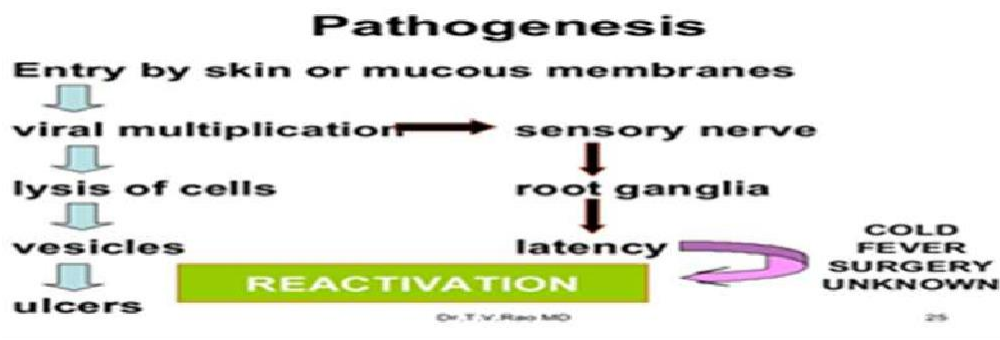
presentations:

- Low grade fever
- Speech disturbance
- Behavioural changes

Herpes Simplex Encephalitis

One of Herpes viruses “Enveloped Double stranded DNA viruses, Characteristic ability to establish **latent infections with periodic reactivation**.”

More serious in immuno-compromised patients



Types:

Herpes simplex 1(above the waist) (Transmitted by contact). Latent in **trigeminal ganglia**.

70% of HSV-1 encephalitis having antibodies suggesting reactivation.

Herpes simplex 2(below the waist)

Transmitted sexually or vertical transmission to newborn “esp. If mother 1ry herpes. Latent in **sacral ganglia**. Can cause neonatal encephalitis “transmitted from mother during delivery”, meningitis, disseminated infection.

Clinical syndromes of HSV

1. Gingivostomatitis
2. Herpes labialis (cold sores or fever blisters)
3. Keratoconjunctivitis
4. Herpetic whitlow
5. Encephalitis
6. Genital herpes
7. Neonatal herpes
8. Disseminated infection in immuno-suppressed pt.

Diagnosis: mostly clinically, IgM antibodies

HSV CSF PCR test is the gold standard method for the diagnosis of HSV encephalitis

MRI

Management and prevention:

Acyclovir “therapy & prophylactic high dose”.

ZIG “Zoster- immune globulins”

Vaccine “live attenuated varicella vaccine

Diagnosis of encephalitis

- Clinical manifestation
- Imaging by MRI, CT EEG
- Laboratory

Diagnosis: CSF

examination

Serological detection of IgM antibodies for HSV, CMV, arboviruses.....

PCR for detection of nucleic acid for viruses

Rabies

Acute viral disease of CNS

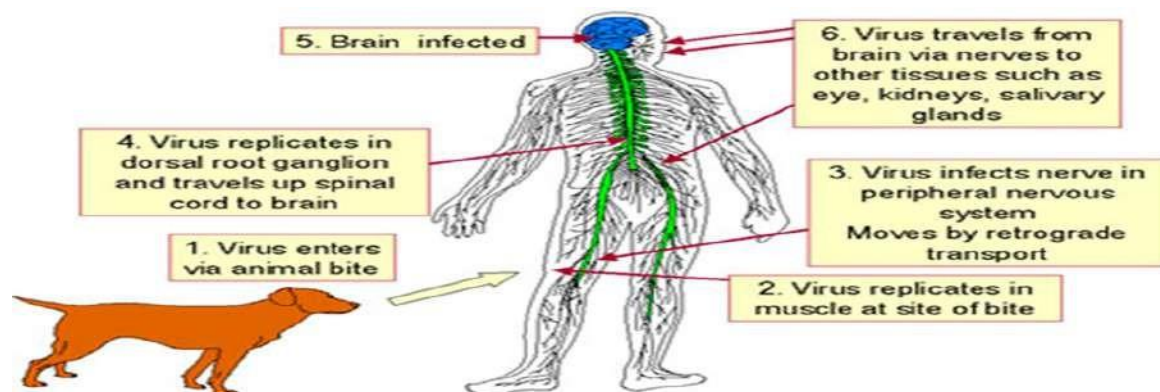
Zoonotic disease

Negative sense-single stranded RNA virus

Secreted in saliva and transmitted by bite or licks of infected animals

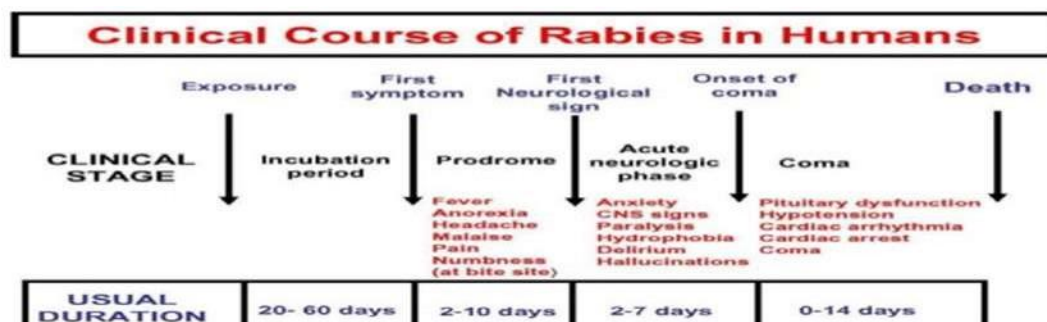
Severity and fatality (if multiple, deep wound, in head and neck)

Mode of infection: by infected saliva through bite or wound



Clinical picture

1. First prodrome (fever, headache, malaise, pain ...)
2. Acute neurological phase: (hydrophobia, anxiety, hallucinations, paralysis, delirium)
3. Coma
4. Death



Based on: Churchill Livingstone Inc.

Diagnosis

Mainly by history taking and clinically

Laboratory in atypical cases

1. Detection of viral antigen in saliva or CSF
2. PCR for detection of RNA
3. Detection of antibodies
4. Post-mortem brain biopsy Negri bodies

Treatment and prevention

Rabies is a preventable disease

Once clinical signs appear, no effective treatment

Post exposure prevention: effective if given before symptoms

Wound cleaning, antibiotic

Passive immunization “Human Rabies Immunoglobulins”

Active immunization “rabies vaccines IM at 0, 3, 7, 14, 28

Preexposure prevention by vaccine to high-risk persons.